

Living Architecture Systems Group

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213 Sterling Road
Toronto, Canada
M6R2B2

2017 IU Summer Camp: Dendrite

April 19th, 2017

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Assembly

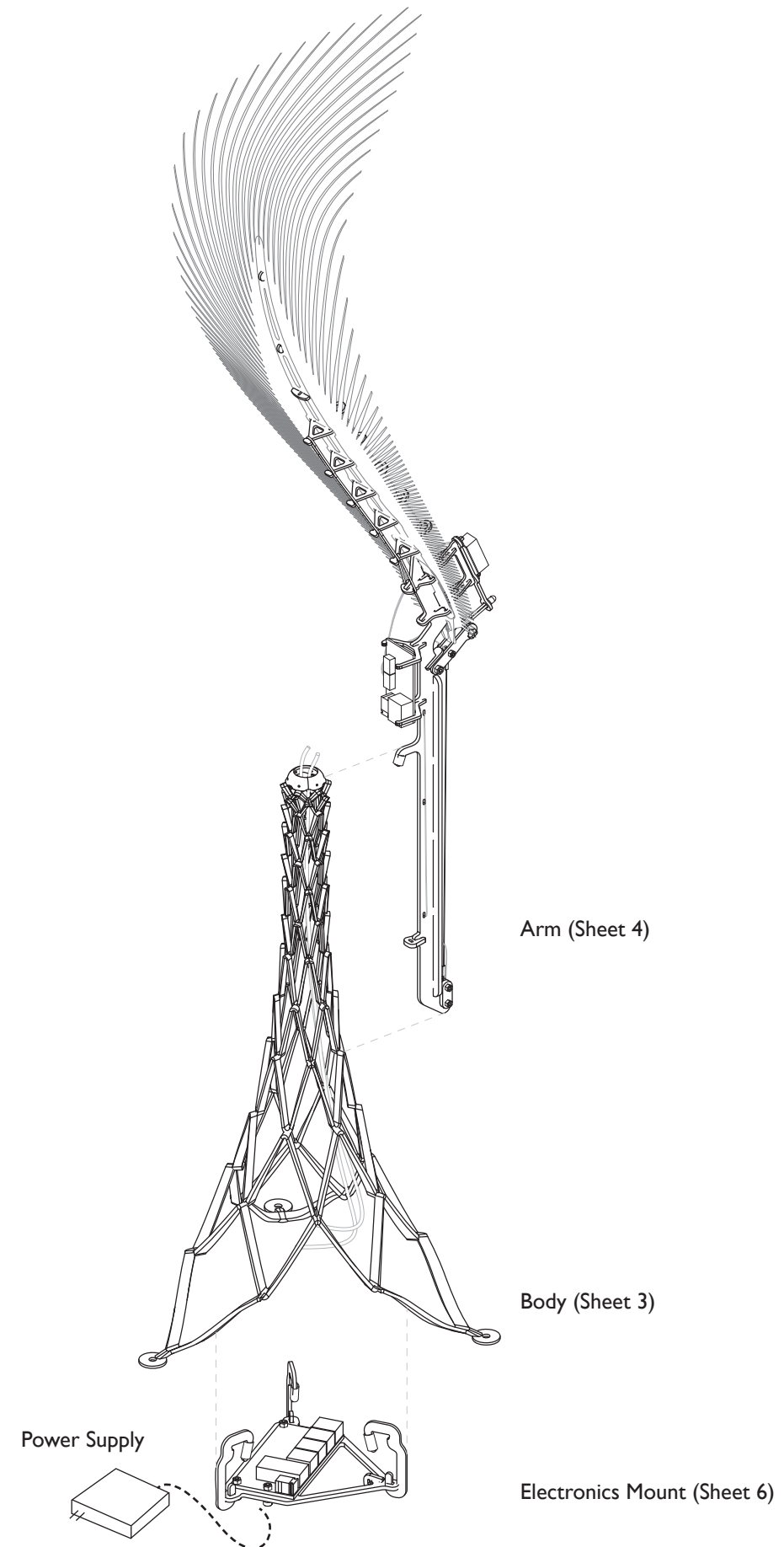
11	Images and Text for Complete Assembly
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1 Dendrite Field: Perspective



2 Dendrite Field: Elevation



3 Assembly Drawing

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Int	Date	Status
JP	17/03/29	Preliminary
JJ	17/04/17	Prototype

Notes:

Status:
Preliminary Draft for
Production

Project:
2017 IU Summer
Camp: Dendrite

Drawing Title:
Grouped Dendrites
and General Assembly

Sheet:
1

- X 1

Monofilament Wire Crimped to Connection Terminal Pin
- X 5

1" Dia Elastic Rubber Band Clear
- X 5

1/8" Dia x 3/8" L Slotted Shoulder Bolt Stainless Steel
- X 1

3/32" Dia x 3/8" L Slotted Shoulder Bolt Stainless Steel
- X 3

2-56 Thread-Forming Flange Nylon Locknut (to match 3/32" Dia Shoulder Bolt and Machine Screw)
- X 5

4-40 Thread-Forming Flange Nylon Locknut (to match 1/8" Dia Shoulder Bolt)
- X 1

Igus iGlide Flange Sleeve Bushing
- X 2

2-56 D x 1/2" L Philips Oval Head Machine Screw
- X 5

6mm OD Black Fishing Round O Ring
- X 1

SMA Wire Assembly
- X 4

2.5mm ID x 5mm OD Silicon Tubing (or equivalent) Cut to 1/2" lengths
- X 2

1/8" ID x 1/4" OD Nylon EVA Tubing Cut to 1/16" lengths
- X 2

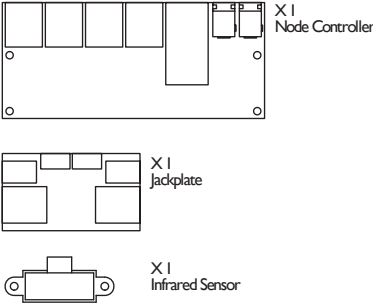
1/16" ID x 3/16" OD Nylon EVA Tubing Cut to 1/16" lengths
- X 1

1/8" ID x 3/8" OD General Purpose Washer
- X 1

T-1 (5mm) Solid State Lamp LED

1

Hardware

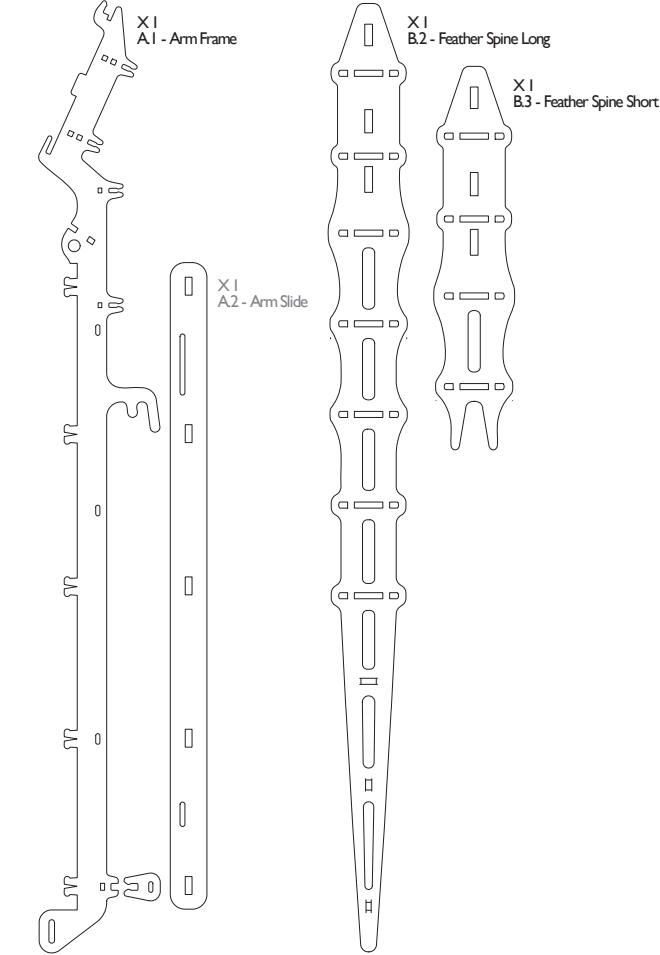


X 1 Jackplate

X 1 Infrared Sensor

4

Circuit Boards and Sensor



X 1 B.2 - Feather Spine Long

X 1 B.3 - Feather Spine Short

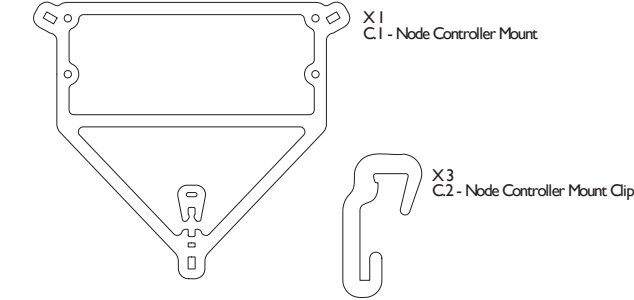
X 1 A.2 - Arm Slide

2

Arm Pieces
Acrylic 3mm

3

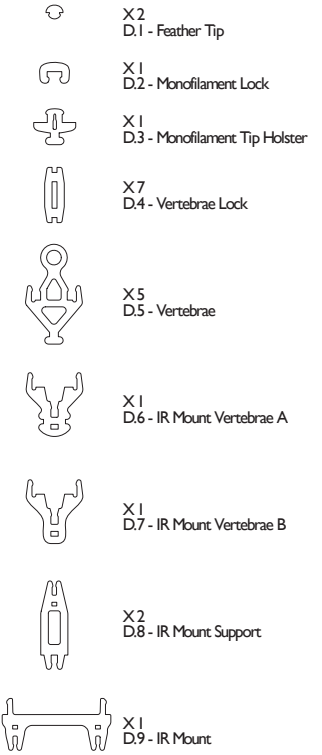
Arm Pieces
Copolyester



X 3 C.2 - Node Controller Mount Clip

5

Circuit Boards Mount



X 1 D.2 - Monofilament Lock

X 1 D.3 - Monofilament Tip Holster

X 7 D.4 - Vertebrae Lock

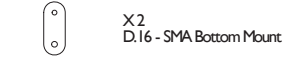
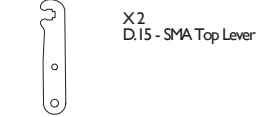
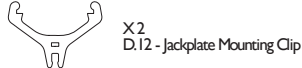
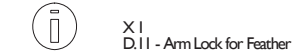
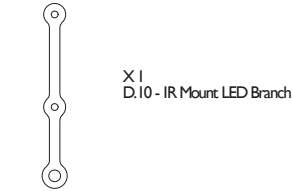
X 5 D.5 - Vertebrae

X 1 D.6 - IR Mount Vertebrae A

X 1 D.7 - IR Mount Vertebrae B

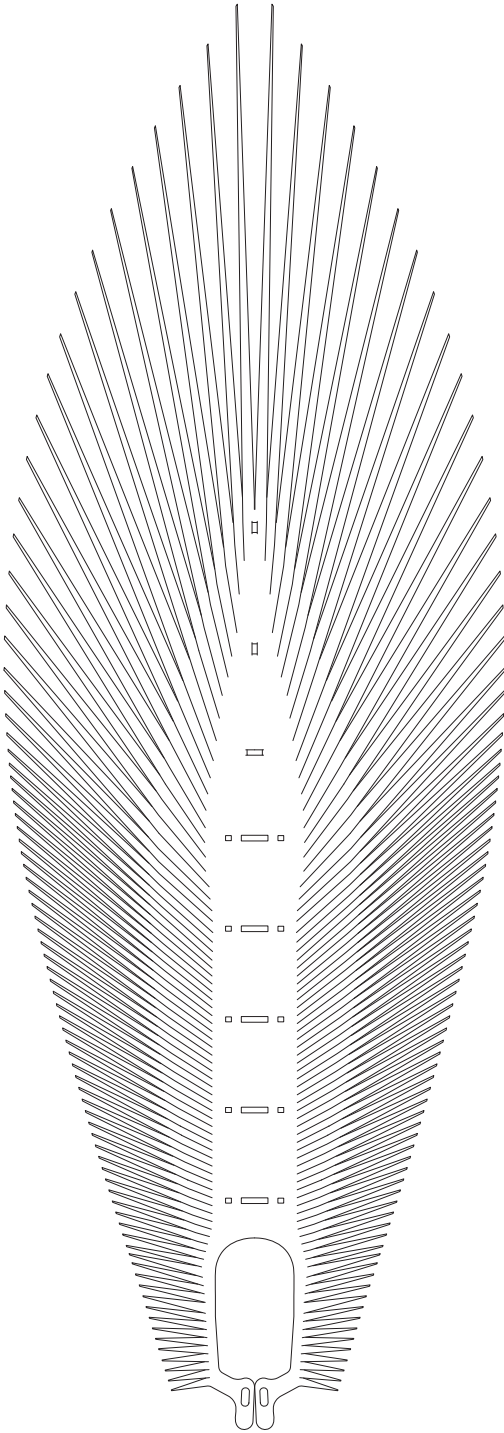
X 2 D.8 - IR Mount Support

X 1 D.9 - IR Mount



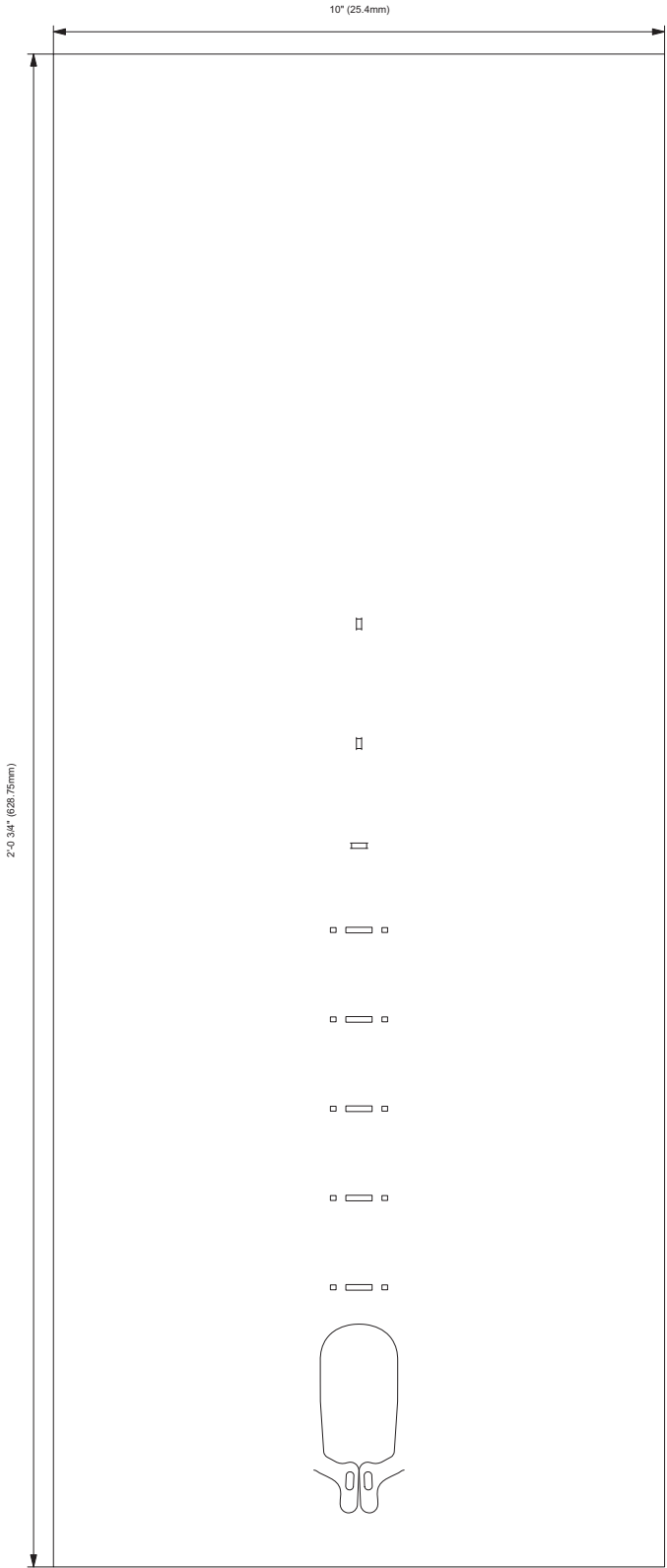
6

Arm Pieces
Acrylic 2mm



7

Feather
Mylar 0.05in



8

Blank Feather Template
Mylar 0.05in

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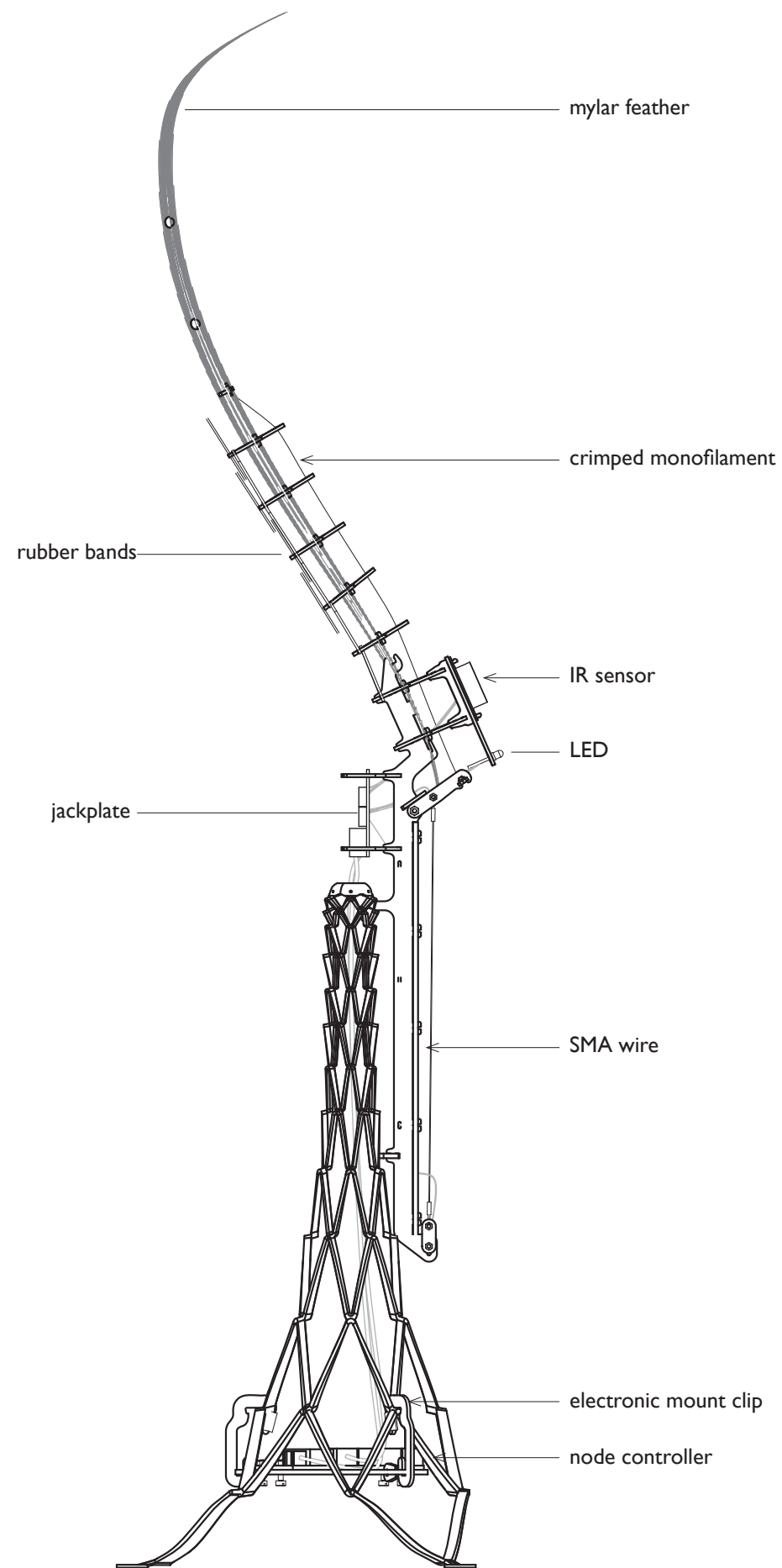
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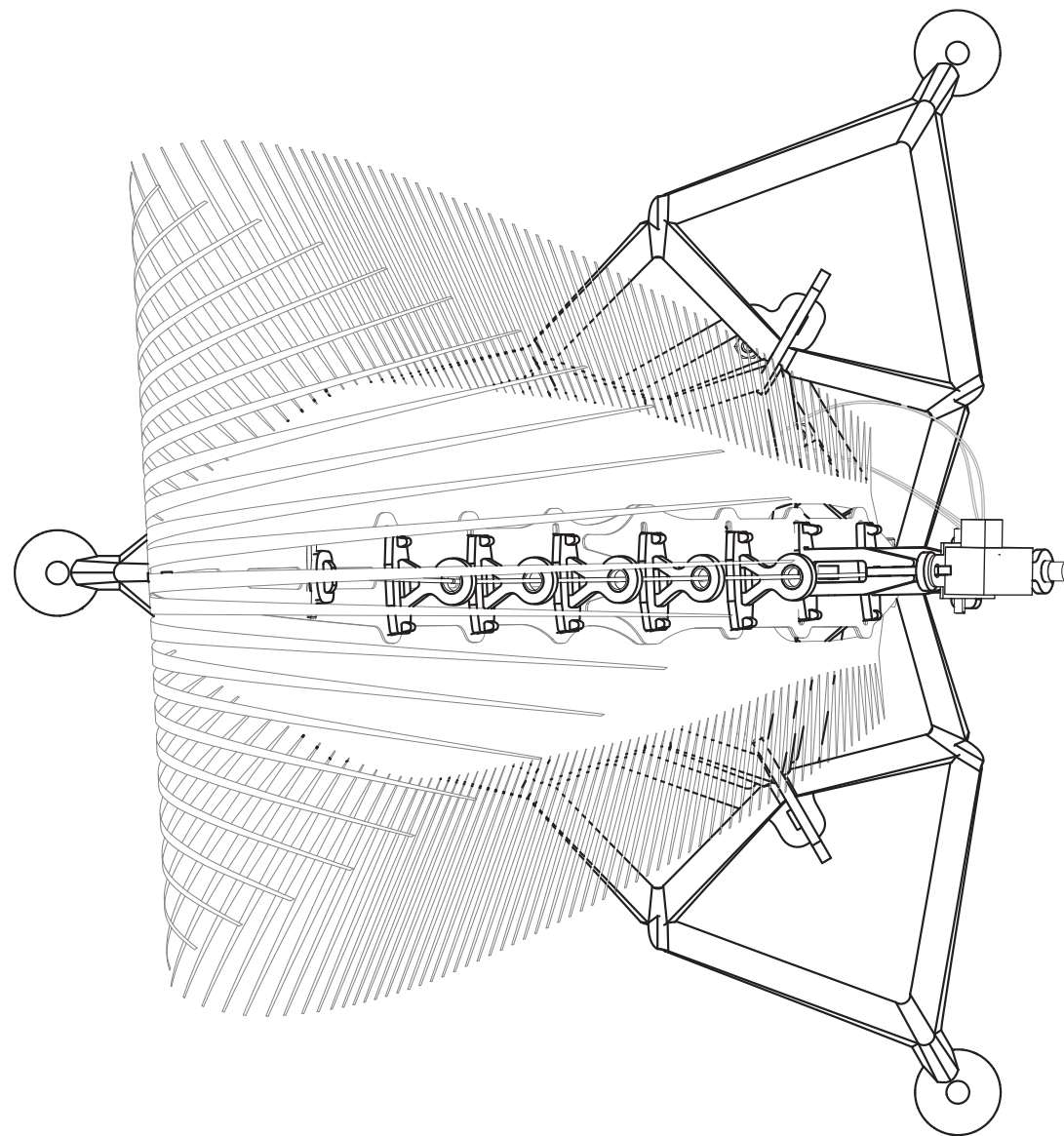
Drawing Title:
Dendrite
Components

Sheet:

2



1 Elevation (Side View)



2 Plan (Top View)

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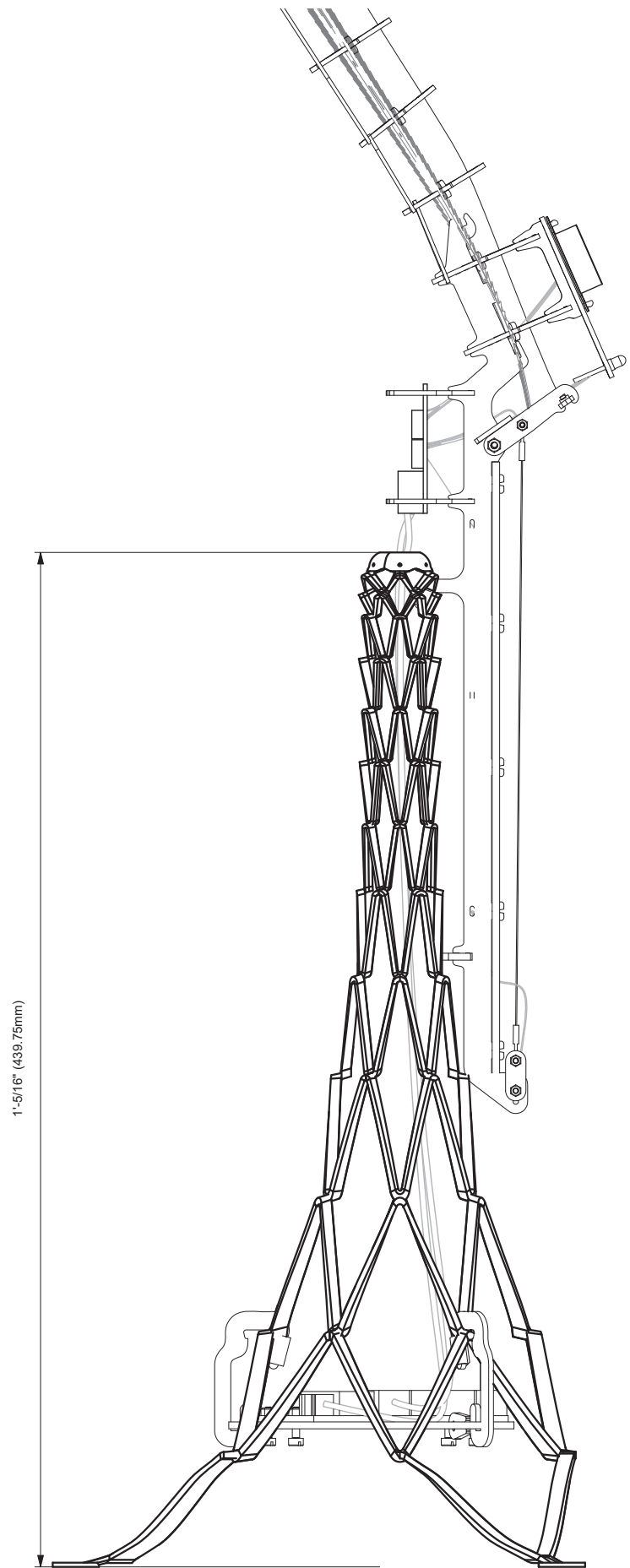
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2017 IU Summer Camp: Dendrite

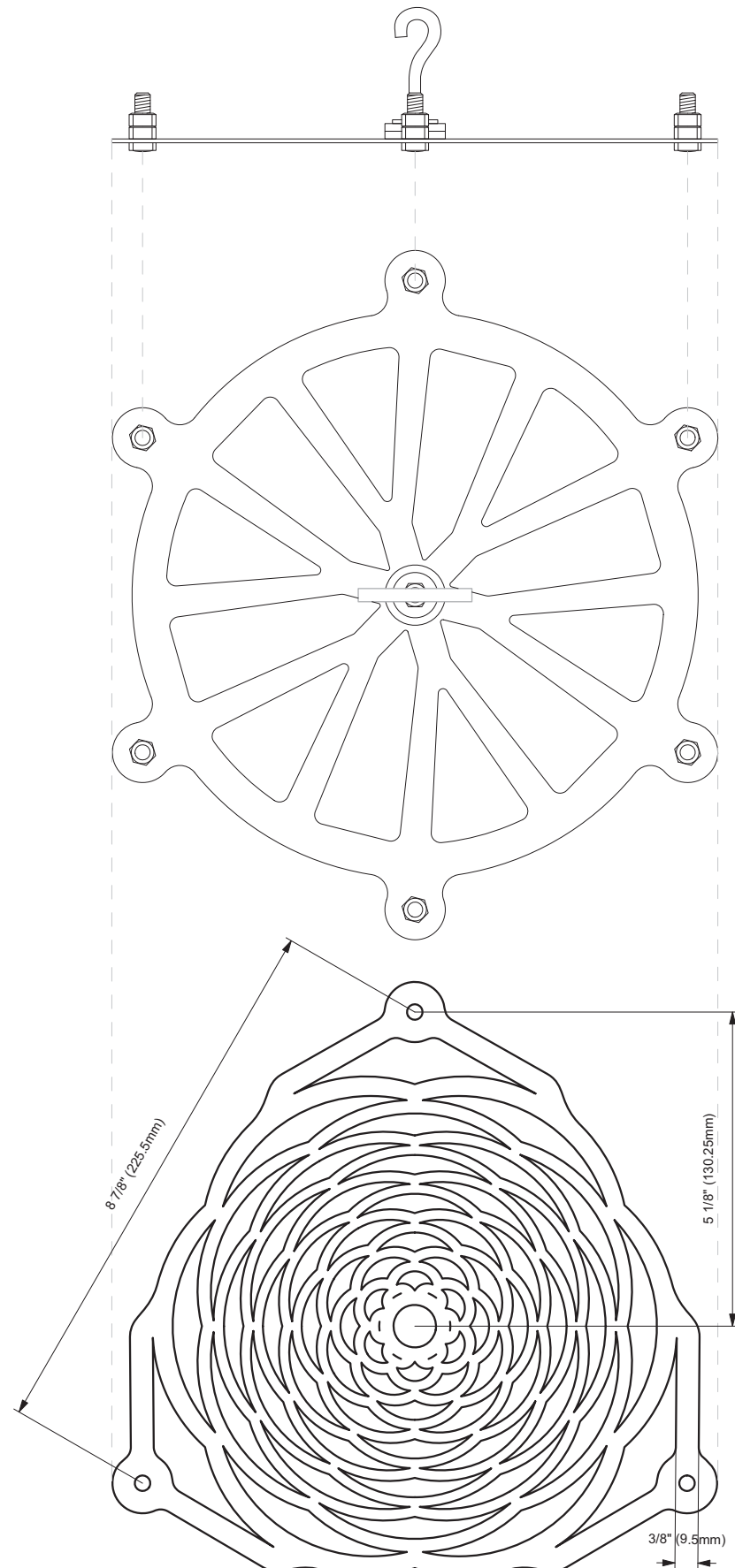
Drawing Title:
Plan (Top View) & Elevation (Side View)

Sheet:

3



1 Elevation
3mm Acrylic



2 Cutting Pattern and Forming Plate
3mm Acrylic Fastened to Steel Frame

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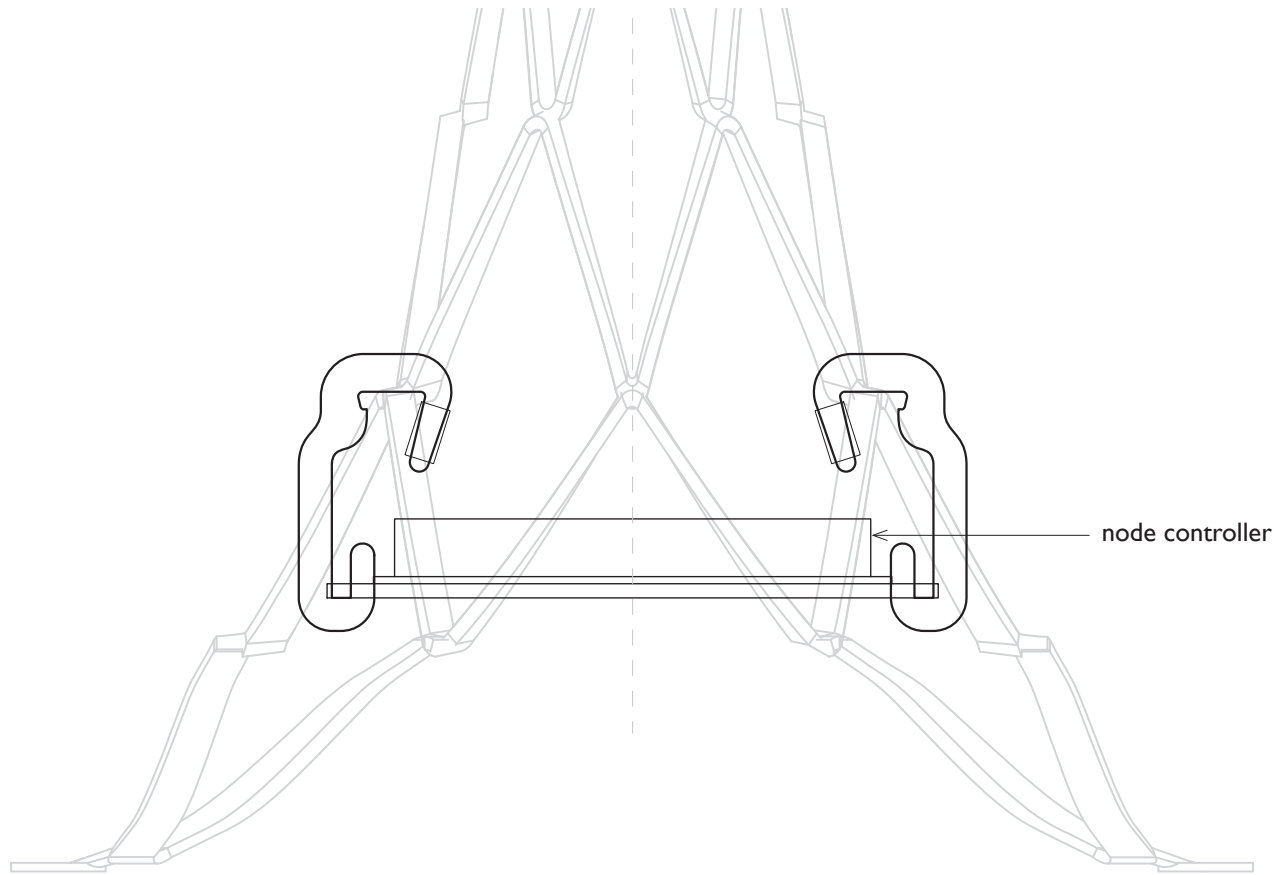
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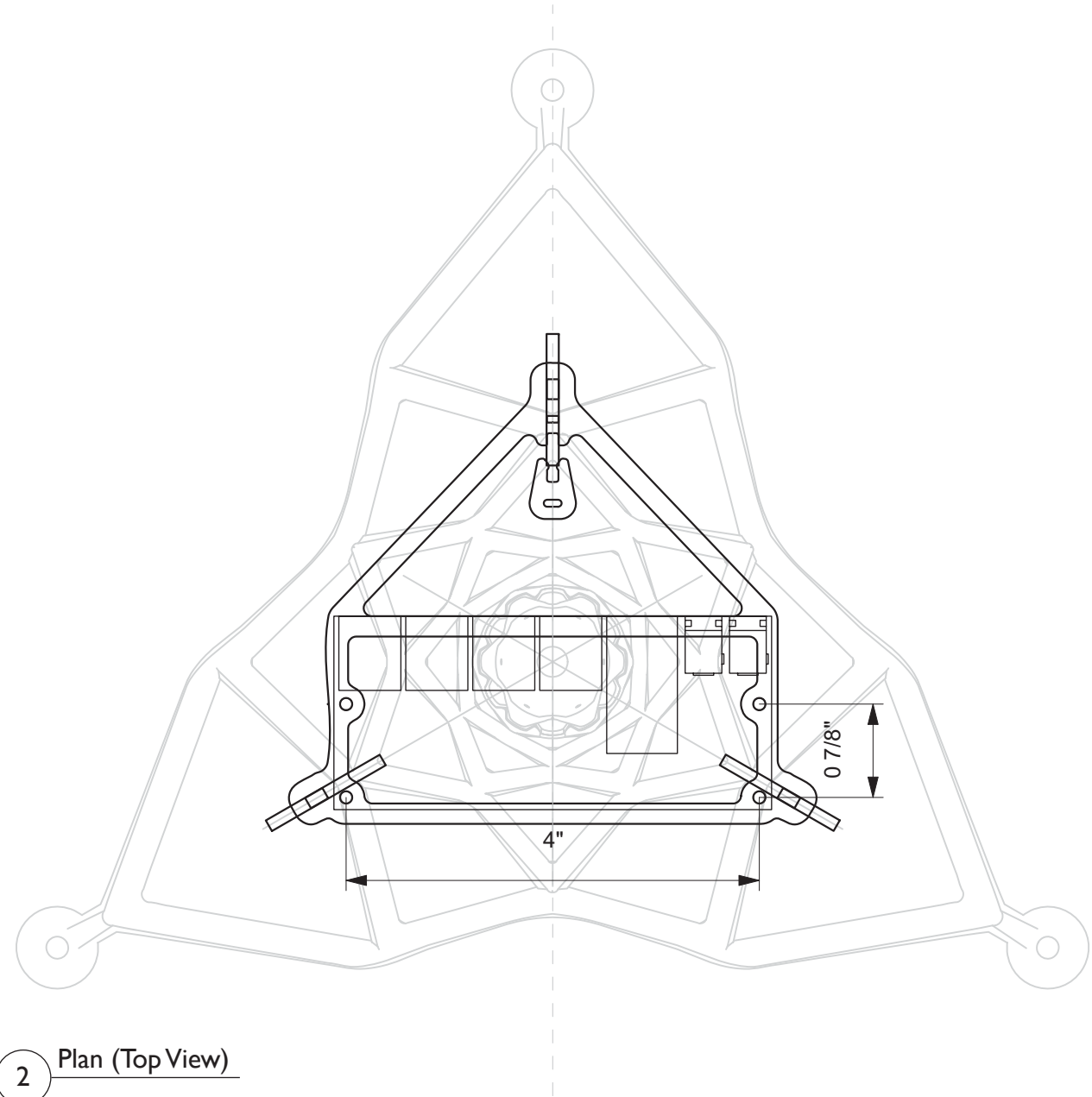
Drawing Title:
Expanded Acrylic Body
with Forming Plate

Sheet:

4



1 Elevation (Front View)



2 Plan (Top View)

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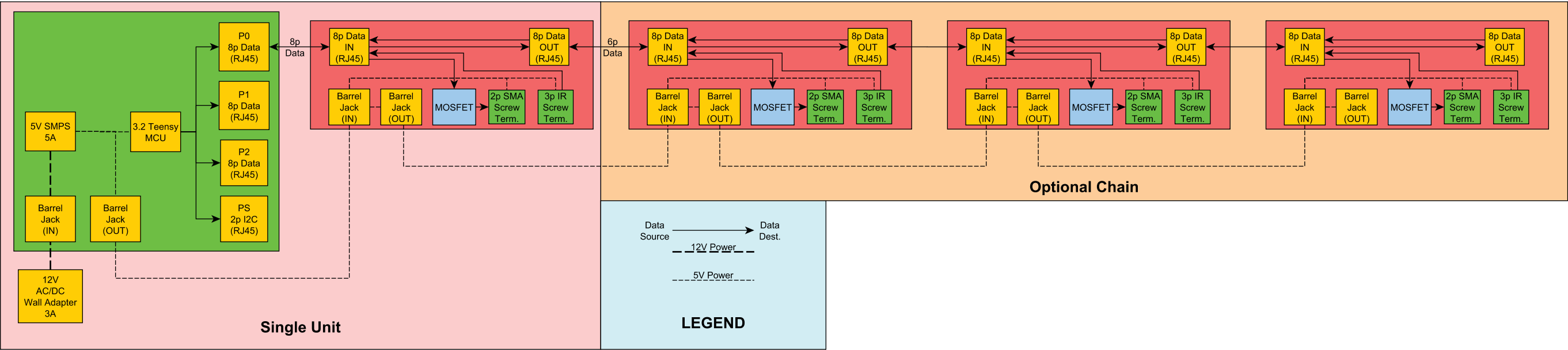
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Drawing Title:
Electronic Mount

Sheet:
5



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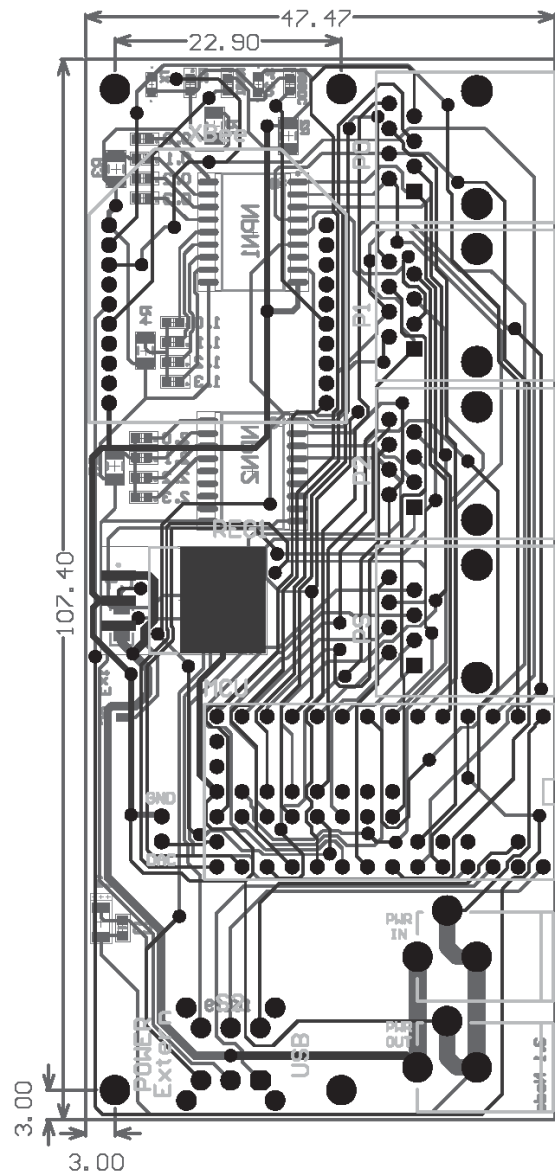
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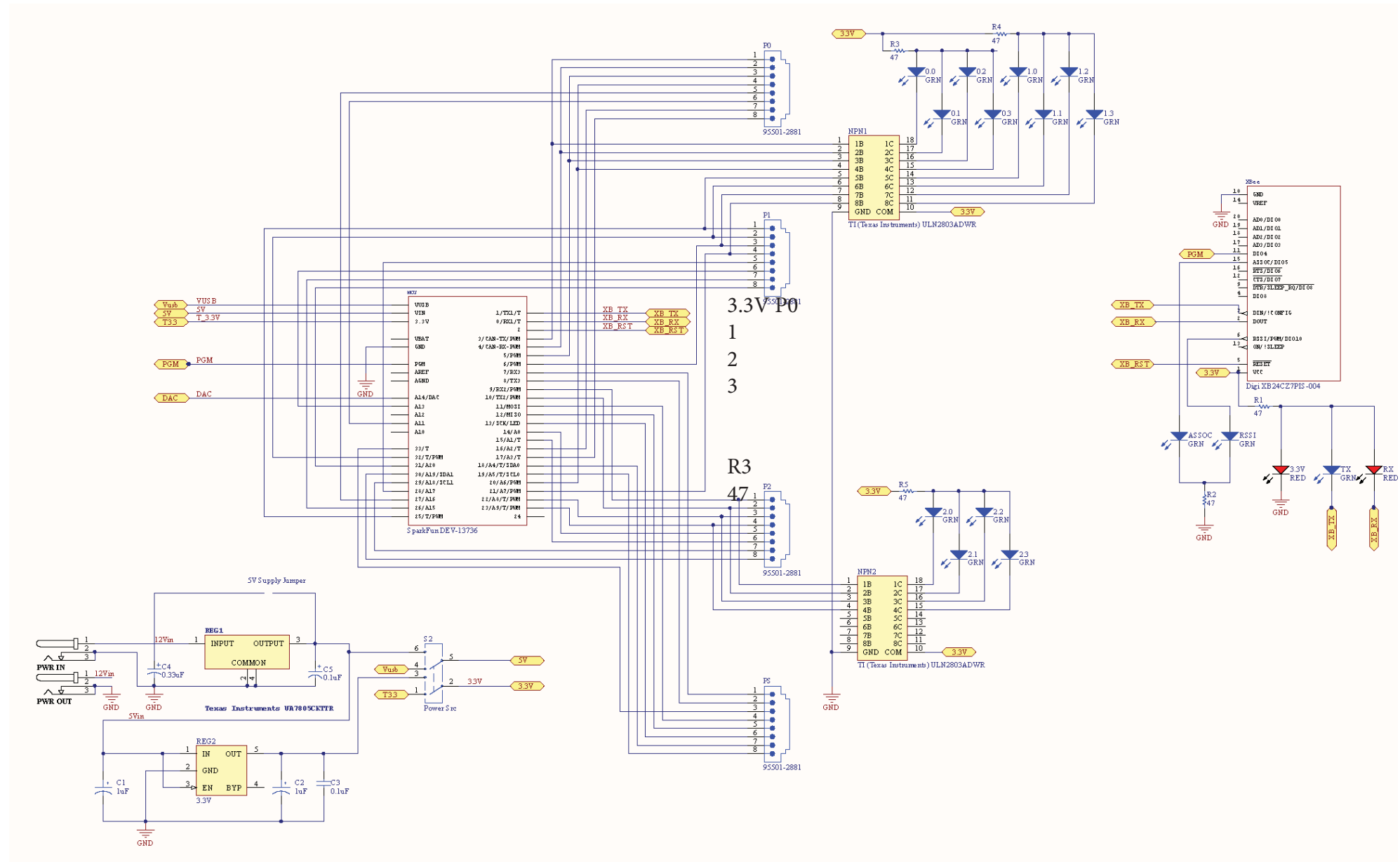
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Drawing Title:
Interactive System
Block Diagram

Sheet:



1 Control Node Detail



2 Control Node Schematic

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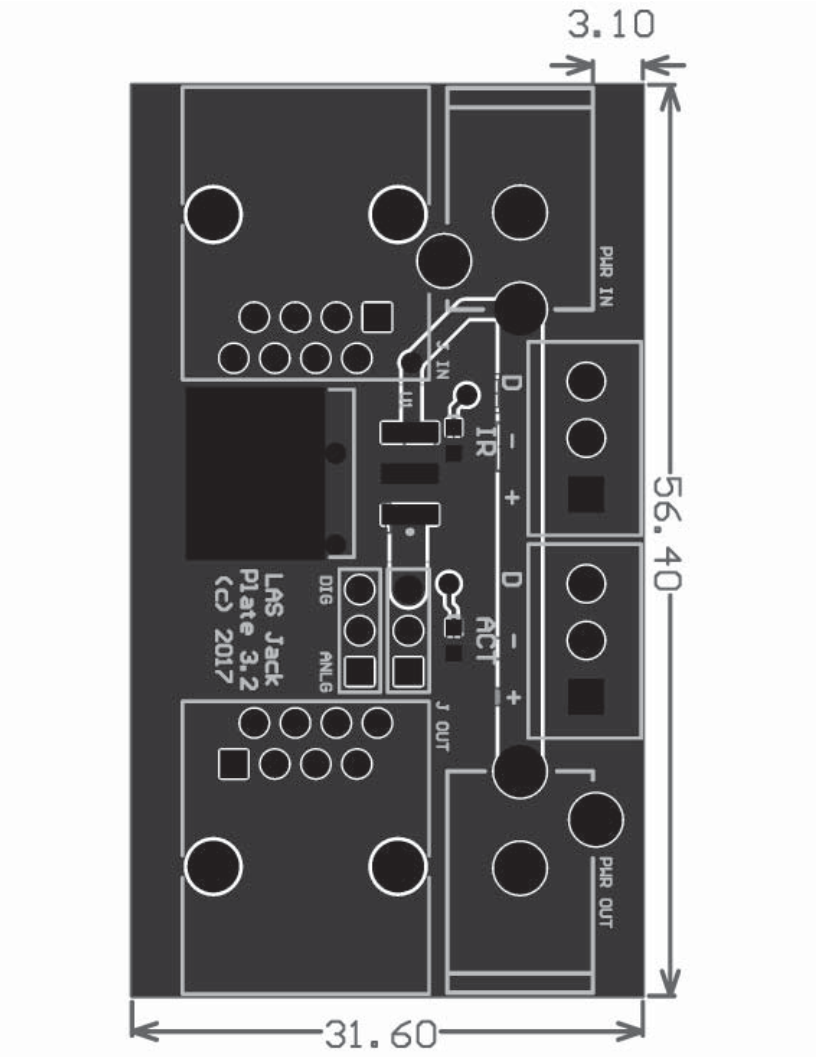
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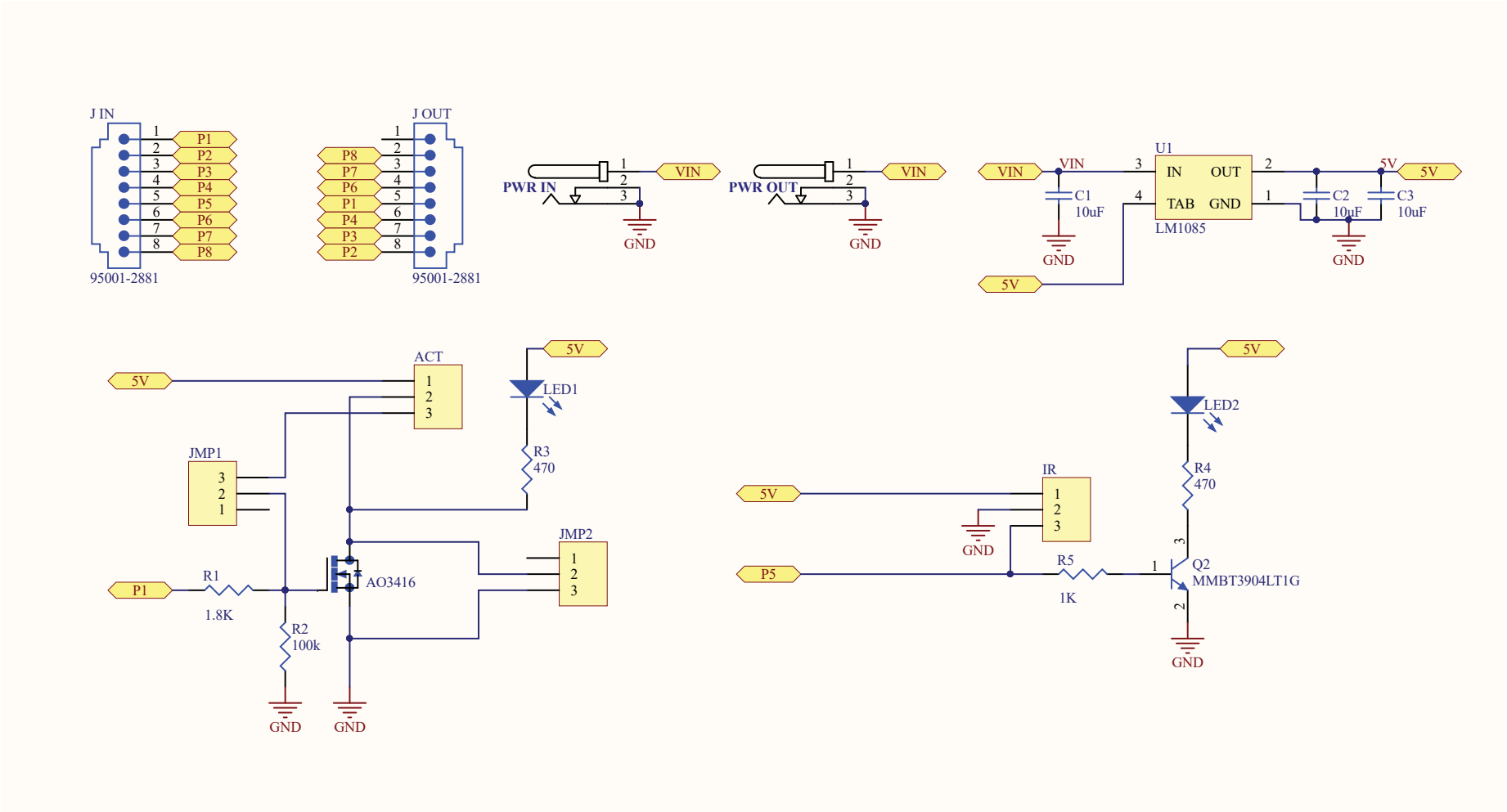
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Drawing Title:
Control Node

Sheet:
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1 Jackplate Detail



2 Jackplate Schematic

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Drawing Title:
Jackplate

Sheet:
8

Total Quantity: 40 IU + 2 PBAI + 2 LASG + 6 Contingent = 50

Electronics Materials								
Type	Name	Provided by PBAI	Unit Cost	Unit Cost	Quantity	Labour	Total (USD)	Notes
Power	12V 3A Wall Adapter - 6' long		\$13.34		50		\$667.00	http://www.mouser.com/ProductDetail/Mean-Well/GST36U12-P11W
Node	Printed Circuit Board (PCB)	✓	\$5.00		50	\$31.20	\$281.20	
	Teensy 3.2 Microcontroller Unit (MCU)	✓	\$19.95		50		\$997.50	https://www.sparkfun.com/products/13736
	5V 3A LM1085 Reg.	✓	\$1.94		50		\$97.00	http://www.digikey.com/product-detail/en/texas-instruments/LM1085ISX-5.0-NOPB/296-35391-1-ND/3739095
	RJ45 Modular Jack	✓	\$0.48		50		\$24.14	http://www.digikey.com/product-detail/en/amphenol-fci/54602-908LF/609-1046-ND/1001360
	3.3V 0.5A MIC5219 Reg.	✓	\$0.76		50		\$38.12	http://www.digikey.com/product-detail/en/microchip-technology/MIC5219-3.3YM5-TR/576-1281-1-ND/771902
	Slide Switch	✓	\$0.57		50		\$28.44	http://www.digikey.com/product-detail/en/c-k/OS202013MT5QN1/CKN9547-ND/1981418
Jack Plate	PCB Board	✓	\$3.00		50	\$16.59	\$166.59	
	RJ45 Modular Jack	✓	\$0.48		50		\$24.14	http://www.digikey.com/product-detail/en/amphenol-fci/54602-908LF/609-1046-ND/1001360
	Metal Oxide Semiconductor Field-Effect Transistor (MOSFET)	✓	\$0.54		50		\$26.85	http://www.digikey.com/product-detail/en/on-semiconductor/NTD5867NLT4G/NTD5867NLT4GOSCT-ND/2409631
	Screw Terminal Blocks	✓	\$0.27		100		\$26.52	http://www.digikey.com/product-detail/en/on-shore-technology-inc/OSTTE020104/ED2740-ND/2351816
Sensors	Infrared (IR) Sensor	*	\$13.25		50		\$662.50	https://www.sparkfun.com/products/242
Actuators	Flexinol Shape-Memory Alloy (SMA)-pre assembled	✓	\$3.00		50		\$150.00	http://www.dynalloy.com/flexwire_70_90.php
	PBAI will give ordering instructions for this							
Connections	Light-Emitting Diode (LED)	✓	\$0.21		50		\$10.50	https://www.digikey.ca/product-detail/en/kingbright/WP7113YD5V/754-1740-ND/3084243
	LED Cabling (24 AWG)	✓	\$0.75		50		\$37.50	http://www.dynalloy.com/crimps.php
	RJ45/Barrel Cabling (20 AWG Barrel, 26 AWG RJ45)	✓	\$0.75		50		\$37.50	https://www.digikey.ca/product-detail/en/tensility-international-corp/30-00399/T1297-30-ND/5023200
	IR Sensor Cable (JST connector pre-assembled)	*	\$1.50		50		\$75.00	https://www.sparkfun.com/products/8733
	Barrel Connectors (2 per cable, 2.1mm ID 5.5mm OD)	✓	\$0.79		100		\$79.00	https://www.digikey.com/product-detail/en/cui-inc/PP3-002A/CP3-1000-ND/992136
	RJ45 Connector (2 per cable)	✓	\$0.23		100		\$23.00	http://www.digikey.com/product-detail/en/Assmann-WSW-Components/A-MO-8-8-F50/AE10316-ND/1957592
Totals							\$3,452.50	Grand Total (40 boards)
							\$86.31	Price of Electronics per Dendrite

Sculptural Materials								
Type	Name		Units per Pkg	Unit Cost/ Pkg Cost	Quantity	Labour	Total (USD)	Notes
Sheet Material								
	Acylic Sheet 0.080" 48"x96" CL EX F2 Acrylite Resist 65			\$98.00	1		\$98.00	specify with manufactuer that large sheets are cut to lasercutter bed dimensions
	Acrylic Sheet 0.118" 48"x96" CL EX F2 Acrylite Resist 65			\$145.00	2		\$290.00	specify with manufactuer that large sheets are cut to lasercutter bed dimensions
	0.8mm or 0.03125" thick PETG Copolyester Plastic Sheet 48"x96"			\$100.00	1		\$100.00	specify with manufactuer that large sheets are cut to lasercutter bed dimensions
	0.005" Grafix Drafting Film (aka Mylar) - double sided Matte ROLL - 36"x60'			\$200.00	1		\$200.00	Cut to fit lasercutter bed dimensions
Hardware								
	Rainbow Loom (or similar) Clear Latex-free color bands (600 band bag)		600 (need 200)	\$3.00	1		\$3.00	
	Nylon Monofilament Wire thickness 0.018" (need a spool of at least 120')			\$10.00	1		\$10.00	
	MOLEX (or similar) Connection Terminal Pin Crimp 0.1" (3mm) barrel diameter			\$0.22	80		\$17.30	
	1/8" D x 3/8" L Slotted Shoulder Bolt Stainless Steel			\$1.33	200		\$266.00	https://www.mcmaster.com/#shoulder-bolts/=171cpce
	3/32" D x 3/8" L Slotted Shoulder Bolt Stainless Steel			\$4.24	80		\$339.20	https://www.mcmaster.com/#shoulder-bolts/=171d35a
	2-56 Thread-Forming Flange Nylon Locknut		100 (need 120)	\$11.91	2		\$23.82	https://www.mcmaster.com/#nylon-hex-locknuts/=171d3nf
	4-40 Thread-Forming Flange Nylon Locknut		200 (need 200)	\$11.91	2		\$23.82	https://www.mcmaster.com/#nylon-hex-locknuts/=171d3nf
	Igus iGlide Flange Sleeve Bushing		50 (need 40)	\$0.54	50		\$27.00	http://www.igus.com/iPro/iPro_02_0001_0000_USen.htm?c=US&l=en
	2-56 D x 1/2" L Philips Oval Head Machine Screw		100 (need 80)	\$5.59	1		\$5.59	https://www.mcmaster.com/#91802a081/=173gzsz
	6mm OD Black Fishing Round O Ring		100 (need 200)	\$1.99	2		\$3.98	provided by PBAI
	2.5mm ID x 5mm OD Flexible Semi-Clear Silicon Tubing							
	(or equivalent) Cut to 1/2" lengths		25' (need 80")	\$0.20	25		\$5.00	
	1/8" ID x 1/4" OD PVC Flexible Semi-Clear Tubing							
	Cut to 1/16" lengths		25' (need 5")	\$0.20	25		\$5.00	https://www.mcmaster.com/#standard-plastic-tubing/=171e7bh
	1/16" ID x 3/16" OD PVC Flexible Semi-Clear or Frosted Tubing							
	Cut to 1/16" lengths		25' (need 5")	\$0.20	25		\$5.00	https://www.mcmaster.com/#standard-plastic-tubing/=171e7bh
	1/8" ID x 3/8" OD General Purpose Metal Washer		100 (need 40)	\$2.04	2		\$4.08	https://www.mcmaster.com/#standard-washers/=171fek2
	White or Clear Self Locking Nylon Cable Cord Wire Zip Ties		100 (need 80)	\$2.50	1		\$2.50	
Totals							\$1,429.29	Grand Total (40 Dendrites)
							\$35.73	Price of Sculptural pieces per Dendrite
							\$122.04	Price Per Dendrite (total)

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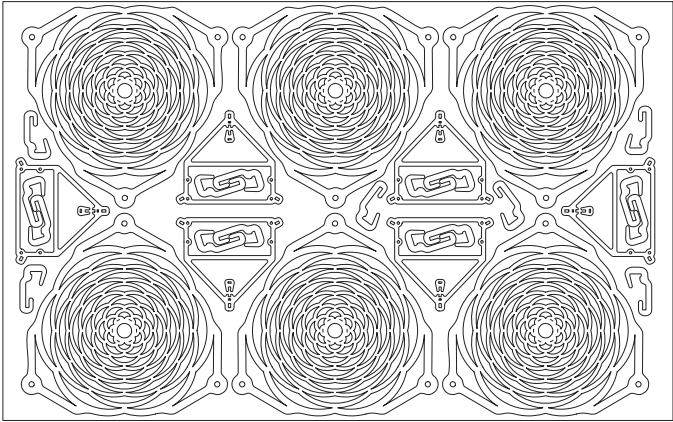
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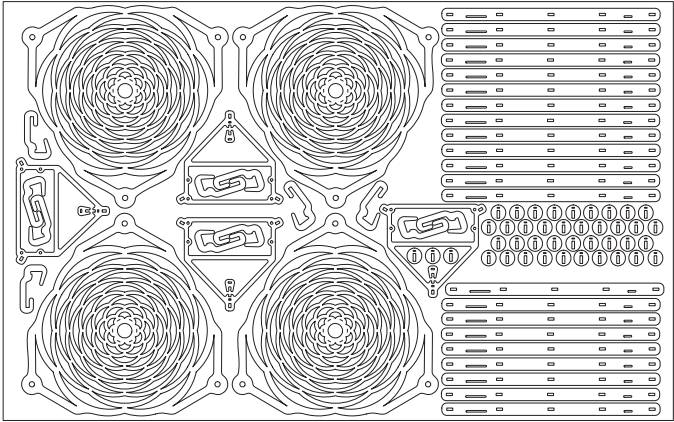
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Camp: Dendrite

Drawing Title:
Bill of Materials

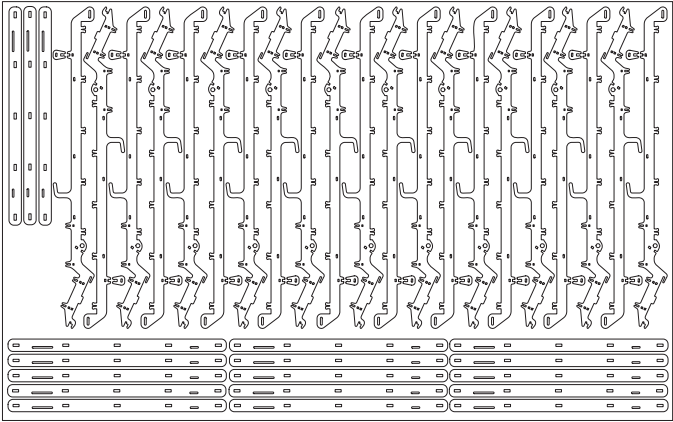
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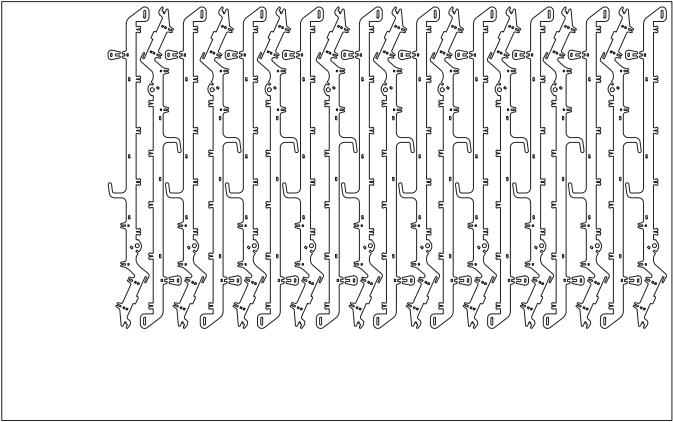
3mm Acrylic x 6



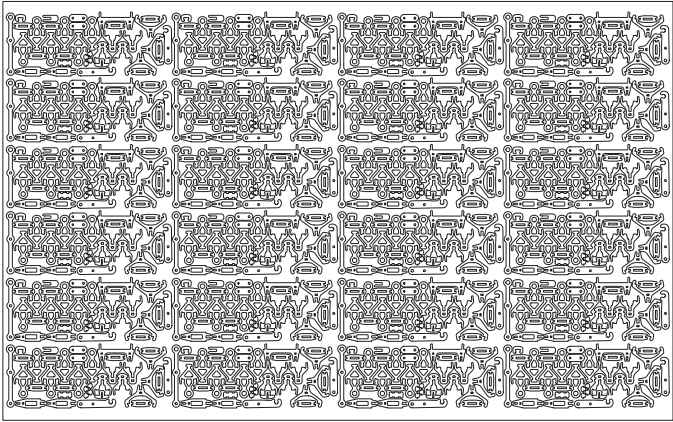
3mm Acrylic x 1



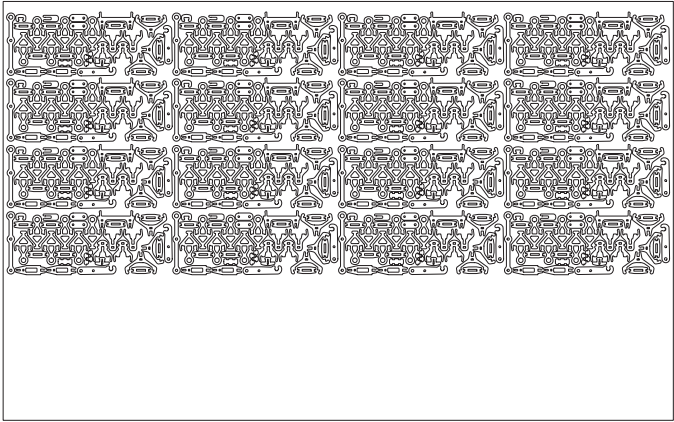
3mm Acrylic x 1



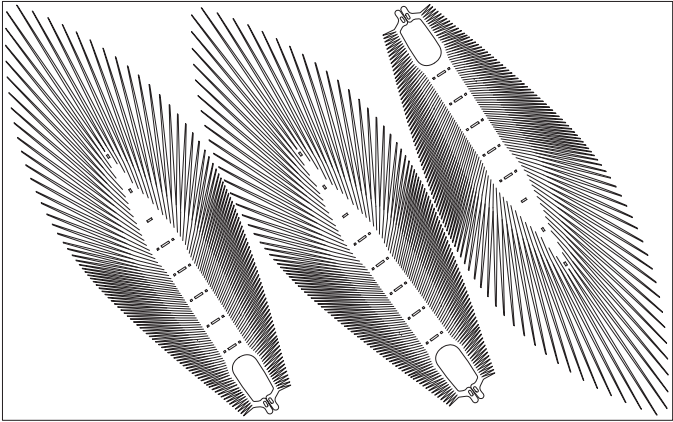
3mm Acrylic x 1



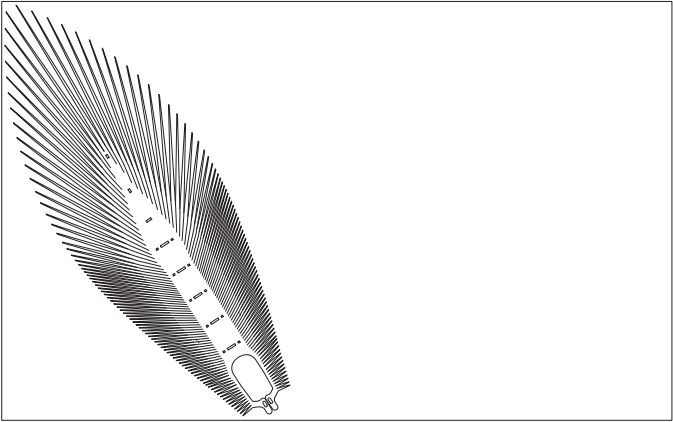
2mm Acrylic x 1



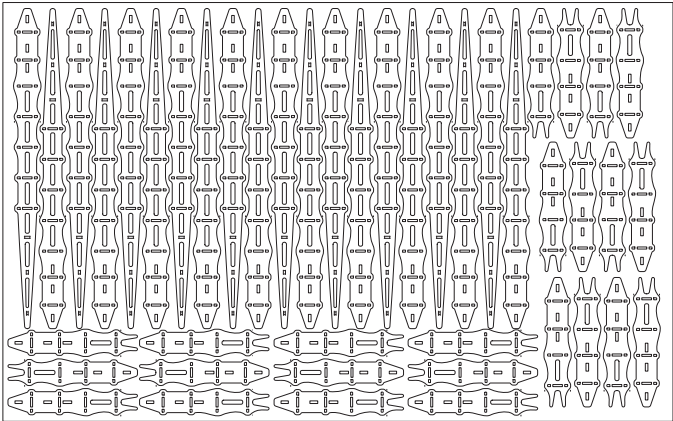
2mm Acrylic x 1



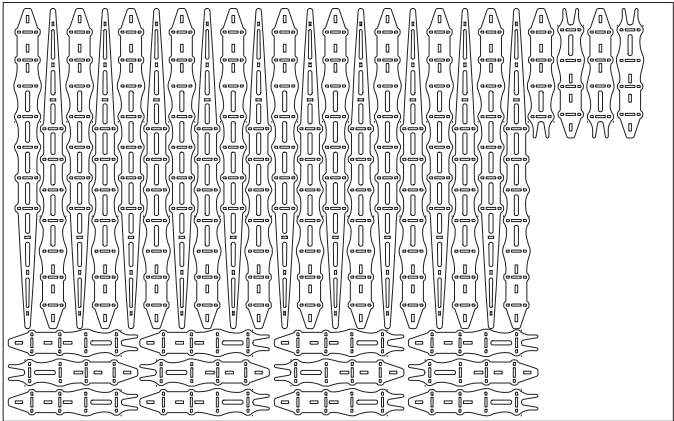
Mylar x 13



Mylar x 13



Copolyester x 1



Copolyester x 1

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